

ELITE IGCSE ACADEMY

Student Past Paper Solutions

Jun 2025 P1H Worked Solutions

Jun 2025 | P1H

31 questions ■ question image followed by worked solution

PREPARED BY

Dr. Eslam Ahmed

Assistant Lecturer, Cairo University Faculty of Engineering

WhatsApp: +20 112 000 9622 | eliteigcse.com

PAST PAPER QUESTION**Q#: 1****Marks: 5****TOPIC: Statistics Toolkit**

Jun 2025 P1H - Jun 2025 | P1H

- 1** The table gives information about the distances 100 adults travel to work.

Distance (d km)	Frequency
$0 < d \leq 5$	26
$5 < d \leq 10$	40
$10 < d \leq 15$	16
$15 < d \leq 20$	10
$20 < d \leq 25$	8

- (a) Write down the modal class.

.....
(1)

- (b) Work out an estimate for the mean distance.

..... km
(4)

(Total for Question 1 is 5 marks)

PAST PAPER SOLUTION

Q#: 1

Marks: 5

TOPIC: **Statistics Toolkit**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 2****Marks: 6****TOPIC: Percentages**

Jun 2025 P1H - Jun 2025 | P1H

- 2** Anna makes cups.
Each cup costs 6 Swiss francs to make.

Anna puts the cups into boxes to sell.
Each box contains 4 cups.

Anna sells 80 boxes of cups for a total of 2160 Swiss francs.

- (a) Work out the percentage profit Anna makes.
Show your working clearly.

..... %
(4)

The height of each cup is 9 cm, correct to the nearest cm

- (b) Write down the lower bound of the height.

..... cm
(1)

The weight of each cup is 120 g, correct to the nearest 10 g

- (c) Write down the upper bound of the weight.

..... g
(1)

(Total for Question 2 is 6 marks)

PAST PAPER SOLUTION

Q#: 2

Marks: 6

TOPIC: **Percentages**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

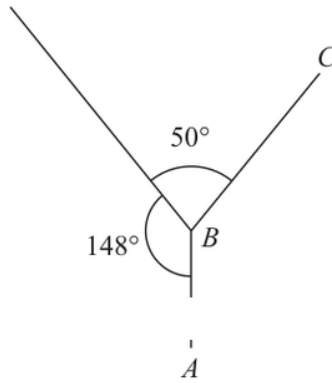
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 3****Marks: 4****TOPIC: Angles in Polygons & Parallel Lines**

Jun 2025 P1H - Jun 2025 | P1H

3Diagram **NOT**
accurately drawn

AB and BC are two sides of a regular polygon with n sides.

Work out the value of n
Show your working clearly.

 $n = \dots\dots\dots$ **(Total for Question 3 is 4 marks)**

PAST PAPER SOLUTION

Q#: 3

Marks: 4

TOPIC: **Angles in Polygons & Parallel Lines**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 4****Marks: 4****TOPIC: Algebraic Roots & Indices**

Jun 2025 P1H - Jun 2025 | P1H

4 $x^5 \times x^7 = x^m$

(a) Find the value of m

$$m = \dots\dots\dots$$

(1)

$y^8 \div y^3 = y^n$

(b) Find the value of n

$$n = \dots\dots\dots$$

(1)

(c) Simplify fully $(5a^4r^2)^3$

$$\dots\dots\dots$$

(2)

(Total for Question 4 is 4 marks)

PAST PAPER SOLUTION

Q#: 4

Marks: 4

TOPIC: **Algebraic Roots & Indices**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 5****Marks: 3****TOPIC: Percentages**

Jun 2025 P1H - Jun 2025 | P1H

- 5** In a sale, normal prices are reduced by 28%
The sale price of a watch is 198 euros.
Work out the normal price of the watch.

..... euros

(Total for Question 5 is 3 marks)

PAST PAPER SOLUTION

Q#: 5

Marks: 3

TOPIC: Percentages

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 6****Marks: 6****TOPIC: Factorising**

Jun 2025 P1H - Jun 2025 | P1H

6 (a) Solve $x - 4 = \frac{3 + 2x}{6}$

Show clear algebraic working.

$$x = \dots\dots\dots (3)$$

(b) (i) Factorise $y^2 - 11y + 30$

$$\dots\dots\dots (2)$$

(ii) Hence solve $y^2 - 11y + 30 = 0$

$$\dots\dots\dots (1)$$

(Total for Question 6 is 6 marks)

PAST PAPER SOLUTION

Q#: 6

Marks: 6

TOPIC: **Factorising**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 7****Marks: 3****TOPIC: Volume & Surface Area**

Jun 2025 P1H - Jun 2025 | P1H

7 Here is a solid cylinder.

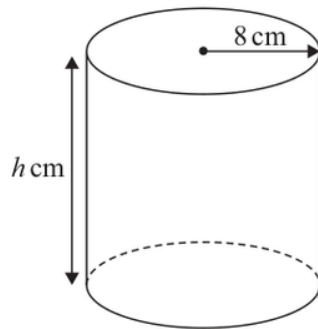


Diagram **NOT**
accurately drawn

The radius of the cylinder is 8 cm

The height of the cylinder is h cm

The volume of the cylinder is 3892 cm^3

Work out the value of h

Give your answer correct to one decimal place.

$h =$

(Total for Question 7 is 3 marks)

PAST PAPER SOLUTION

Q#: 7

Marks: 3

TOPIC: **Volume & Surface Area**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 8****Marks: 4****TOPIC: Powers, Roots & Standard Form**

Jun 2025 P1H - Jun 2025 | P1H

8 (a) Write 520 million in standard form.

.....
(1)

(b) Write 8.79×10^{-5} as an ordinary number.

.....
(1)

(c) Work out $(5 \times 10^{42}) \times (7 \times 10^{-180})$
Give your answer in standard form.

.....
(2)

(Total for Question 8 is 4 marks)

PAST PAPER SOLUTION

Q#: 8

Marks: 4

TOPIC: **Powers, Roots & Standard Form**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

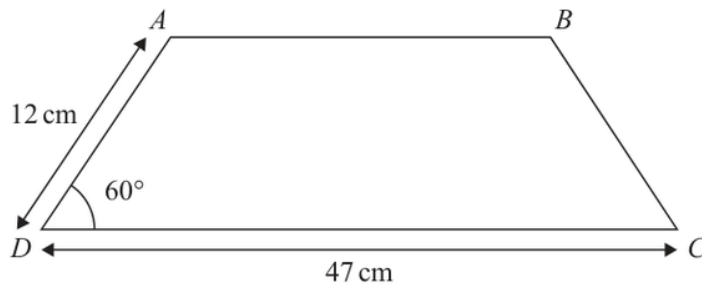
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 9****Marks: 5****TOPIC: Area & Perimeter**

Jun 2025 P1H - Jun 2025 | P1H

9Diagram **NOT**
accurately drawn*ABCD* is a trapezium with one line of symmetry.

$$\text{angle } ADC = 60^\circ \quad AD = 12 \text{ cm} \quad DC = 47 \text{ cm}$$

Work out the area of the trapezium.

Give your answer correct to 3 significant figures.

Show your working clearly.

..... cm²

(Total for Question 9 is 5 marks)

PAST PAPER SOLUTION

Q#: 9

Marks: 5

TOPIC: **Area & Perimeter**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION

Q#: 10

Marks: 3

TOPIC: **Graphs of Functions**

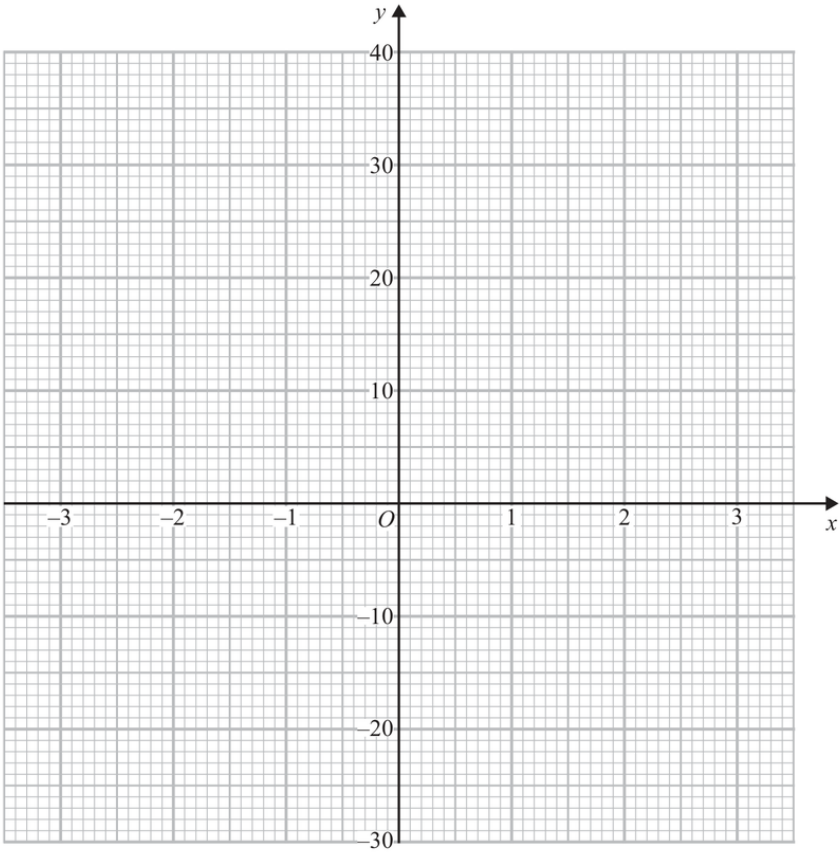
Jun 2025 P1H - Jun 2025 | P1H

10 (a) Complete the table of values for $y = x^3 + 2x + 3$

x	-3	-2	-1	0	1	2	3
y		-9	0	3	6		36

(1)

(b) On the grid, draw the graph of $y = x^3 + 2x + 3$ for $-3 \leq x \leq 3$



(2)

(Total for Question 10 is 3 marks)

PAST PAPER SOLUTION

Q#: 10

Marks: 3

TOPIC: **Graphs of Functions**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 11****Marks: 2****TOPIC: Circles, Arcs & Sectors**

Jun 2025 P1H - Jun 2025 | P1H

- 11** $OABC$ is a sector of a circle, centre O

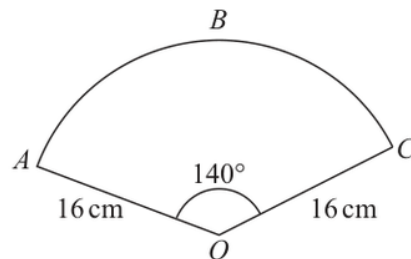


Diagram **NOT**
accurately drawn

$$\text{angle } AOC = 140^\circ$$

$$OA = OC = 16 \text{ cm}$$

Calculate the area of the sector.

Give your answer correct to 3 significant figures.

..... cm^2

(Total for Question 11 is 2 marks)

PAST PAPER SOLUTION

Q#: 11

Marks: 2

TOPIC: **Circles, Arcs & Sectors**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 12****Marks: 3****TOPIC: Algebraic Fractions**

Jun 2025 P1H - Jun 2025 | P1H

12 Express $\frac{5}{4} + \frac{x-3}{6x}$ as a single fraction in its simplest form.

(Total for Question 12 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 12

Marks: 3

TOPIC: **Algebraic Fractions**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 13****Marks: 3****TOPIC: Compound Interest & Depreciation**

Jun 2025 P1H - Jun 2025 | P1H

13 Charlotte bought an apartment for \$750 000

In the first year, the value of the apartment decreased by 4%

In the second year, the value of the apartment decreased by 6.5%

In the third year, the value of the apartment increased by $x\%$

At the end of the third year, the value of Charlotte's apartment was \$698 445

Work out the value of x $x = \dots\dots\dots$ **(Total for Question 13 is 3 marks)**

EA

PAST PAPER SOLUTION

Q#: 13

Marks: 3

TOPIC: **Compound Interest & Depreciation**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 14****Marks: 4****TOPIC: Probability Diagrams - Venn & Tree Diagrams**

Jun 2025 P1H - Jun 2025 | P1H

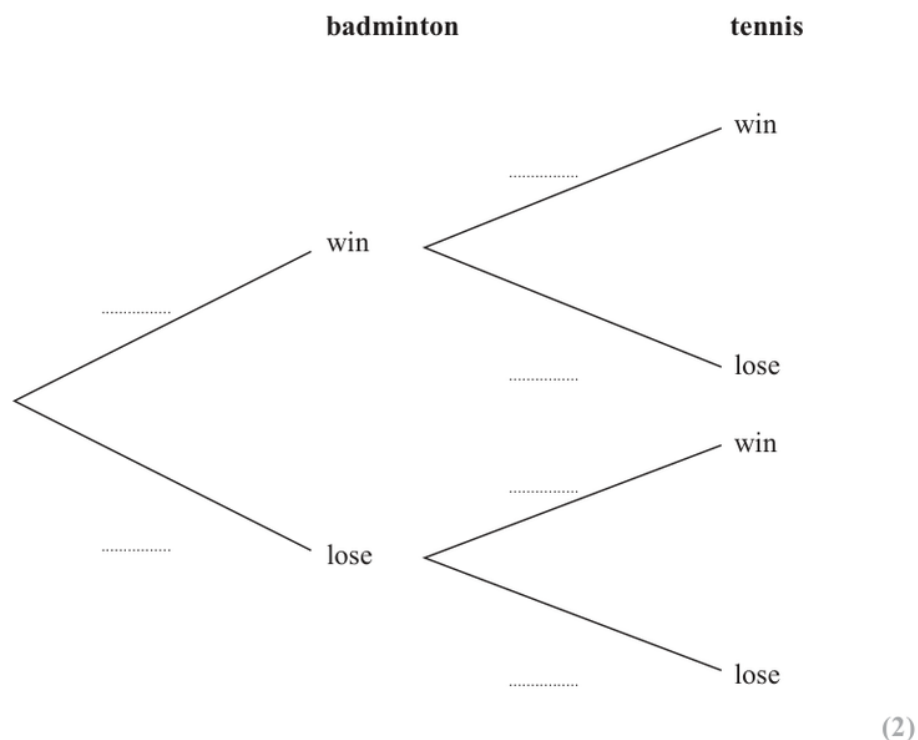
14 Ricardo is going to play one game of badminton and one game of tennis.

He can either win or lose each game.

The probability that he will win the game of badminton is 0.7

The probability that he will win the game of tennis is 0.4

(a) Complete the probability tree diagram.



(b) Work out the probability that Ricardo loses both games.

.....
(2)

(Total for Question 14 is 4 marks)

PAST PAPER SOLUTION

Q#: 14

Marks: 4

TOPIC: **Probability Diagrams - Venn & Tree Diagrams**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 15****Marks: 4****TOPIC: Algebraic Proof**

Jun 2025 P1H - Jun 2025 | P1H

15 $(\sqrt{3})^5 = k\sqrt{3}$ where k is an integer.

(a) Find the value of k

$$k = \dots\dots\dots (1)$$

(b) Show that $\frac{21}{3 - \sqrt{2}}$ can be written in the form $c + \sqrt{d}$

where c and d are integers.

Show each stage of your working clearly.

(3)

(Total for Question 15 is 4 marks)

PAST PAPER SOLUTION

Q#: 15

Marks: 4

TOPIC: **Algebraic Proof**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 16****Marks: 4****TOPIC: Functions**

Jun 2025 P1H - Jun 2025 | P1H

16 $f(x) = 9 - \sqrt{x}$ where $x \geq 0$
 $g(x) = 4x^2$

(a) Find $f(9)$

.....
(1)

(b) Solve $fg(x) < 0$
Show clear algebraic working.

.....
(3)

.....
(Total for Question 16 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 16

Marks: 4

TOPIC: **Functions**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

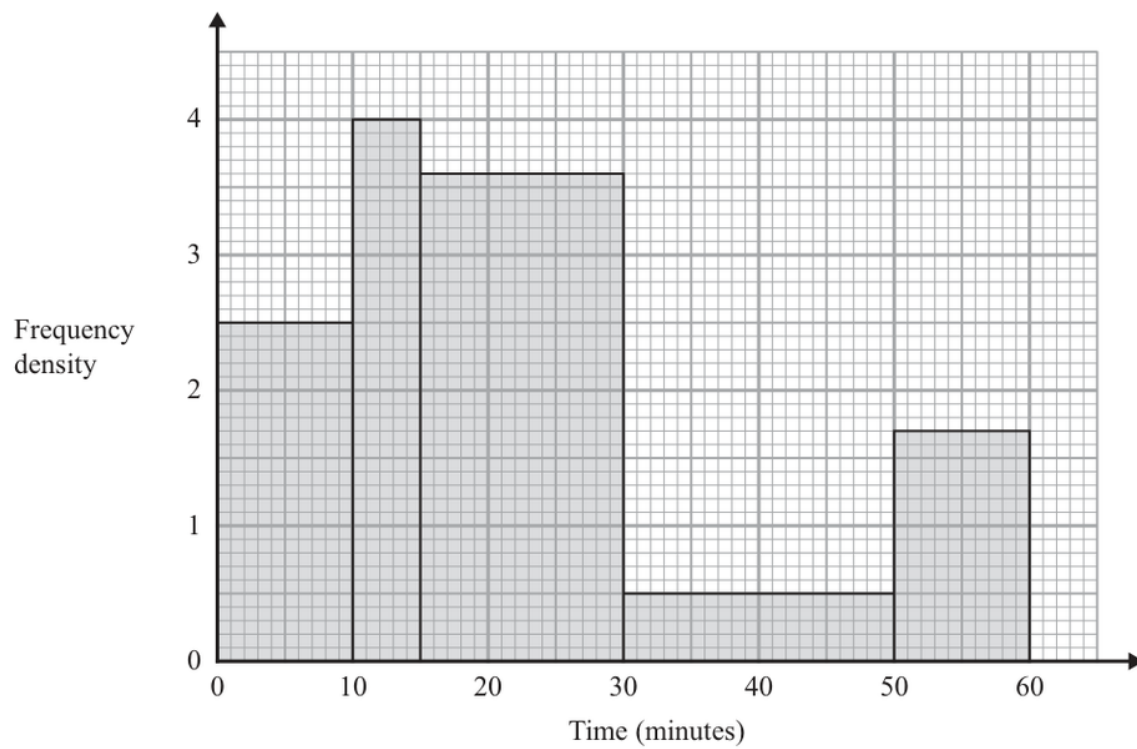
No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 17****Marks: 3****TOPIC: Histograms**

Jun 2025 P1H - Jun 2025 | P1H

- 17 The histogram shows information about the times, in minutes, that some people were in a shop.



Work out an estimate for the proportion of these people who were in the shop for more than 40 minutes.

(Total for Question 17 is 3 marks)

PAST PAPER SOLUTION

Q#: 17

Marks: 3

TOPIC: **Histograms**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 18****Marks: 4****TOPIC: Direct & Inverse Proportion**

Jun 2025 P1H - Jun 2025 | P1H

18 y is inversely proportional to x^3

$$y = 4 \text{ when } x = 3$$

Work out the value of x when $y = 864$

$$x = \dots\dots\dots$$

(Total for Question 18 is 4 marks)

EA

PAST PAPER SOLUTION

Q#: 18

Marks: 4

TOPIC: **Direct & Inverse Proportion**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 19****Marks: 2****TOPIC: Transformations of Graphs**

Jun 2025 P1H - Jun 2025 | P1H

19 The equation of a curve is $y = f(x)$

There is only one minimum point on the curve.

The coordinates of this minimum point are $(8, -12)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x) + 3$

(..... ,)
(1)

(ii) $y = f(2x)$

(..... ,)
(1)**(Total for Question 19 is 2 marks)**

PAST PAPER SOLUTION

Q#: 19

Marks: 2

TOPIC: **Transformations of Graphs**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

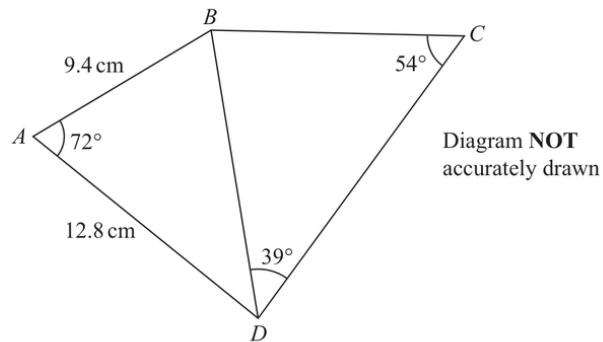
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jun 2025 P1H - Jun 2025 | P1H

20

Work out the length of BC
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20

Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

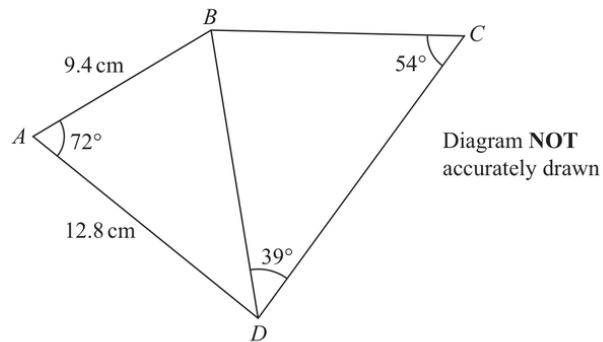
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 20****Marks: 5****TOPIC: Sine, Cosine Rule & Area of Triangles**

Jun 2025 P1H - Jun 2025 | P1H

20

Work out the length of BC
Give your answer correct to 3 significant figures.

..... cm

(Total for Question 20 is 5 marks)

PAST PAPER SOLUTION

Q#: 20

Marks: 5

TOPIC: **Sine, Cosine Rule & Area of Triangles**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Probability Toolkit**

Jun 2025 P1H - Jun 2025 | P1H

21 A box contains 20 counters.

9 of the counters are red

7 of the counters are yellow

4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

(Total for Question 21 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 3

TOPIC: **Probability Toolkit**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 21****Marks: 3****TOPIC: Probability Toolkit**

Jun 2025 P1H - Jun 2025 | P1H

21 A box contains 20 counters.

9 of the counters are red

7 of the counters are yellow

4 of the counters are green

Alex takes at random three counters from the box.

Work out the probability that exactly two of the three counters are the same colour.

(Total for Question 21 is 3 marks)

EA

PAST PAPER SOLUTION

Q#: 21

Marks: 3

TOPIC: **Probability Toolkit**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

Jun 2025 P1H - Jun 2025 | P1H

22 Solve the simultaneous equations

$$\begin{aligned}x^2 + 3y + y^2 &= 7 \\ y &= x + 2\end{aligned}$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 22****Marks: 5****TOPIC: Simultaneous Equations**

Jun 2025 P1H - Jun 2025 | P1H

22 Solve the simultaneous equations

$$\begin{aligned}x^2 + 3y + y^2 &= 7 \\ y &= x + 2\end{aligned}$$

Show clear algebraic working.

(Total for Question 22 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 22

Marks: 5

TOPIC: **Simultaneous Equations**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

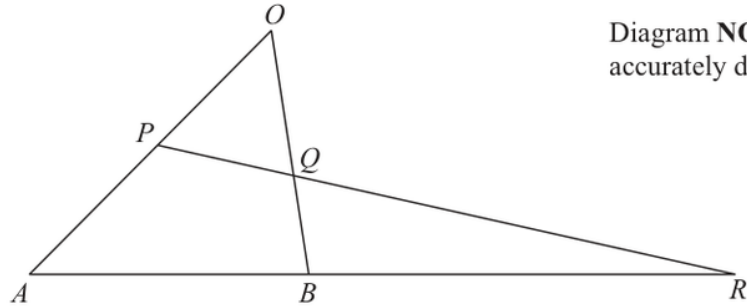
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 23****Marks: 5**TOPIC: **Vectors**

Jun 2025 P1H - Jun 2025 | P1H

23Diagram **NOT**
accurately drawn OAB is a triangle. P is the midpoint of OA Q is a point on OB ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

(a) Express $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
(1)

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

(b) Use a vector method to find the value of n

$$n = \text{.....}$$

(4)

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 5

TOPIC: **Vectors**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

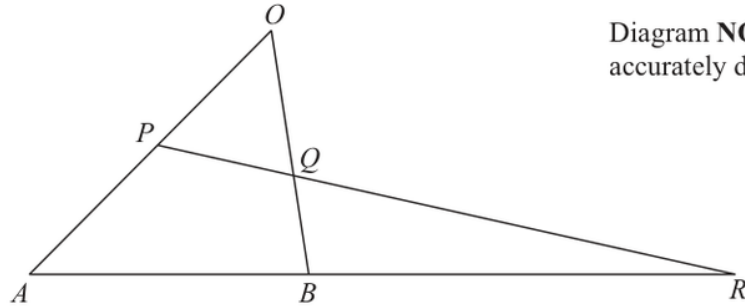
Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 23****Marks: 5**TOPIC: **Vectors**

Jun 2025 P1H - Jun 2025 | P1H

23Diagram **NOT**
accurately drawn OAB is a triangle. P is the midpoint of OA Q is a point on OB ABR and PQR are straight lines.

$$\vec{OA} = 12\mathbf{a} \quad \vec{OB} = 8\mathbf{b}$$

(a) Express $\vec{AB}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
(1)

$$AB:BR = 1:2 \quad \vec{OQ} = n\mathbf{b}$$

(b) Use a vector method to find the value of n

$$n = \text{.....}$$

(4)

(Total for Question 23 is 5 marks)

PAST PAPER SOLUTION

Q#: 23

Marks: 5

TOPIC: **Vectors**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Sequences**

Jun 2025 P1H - Jun 2025 | P1H

24 An arithmetic series has 30 terms.

The first term is a

The common difference is d

The 20th term is 123

The sum of the 30 terms is 2880

Work out the value of a and the value of d
Show clear algebraic working.

 $a = \dots\dots\dots$ $d = \dots\dots\dots$

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24

Marks: 5

TOPIC: **Sequences**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 24****Marks: 5****TOPIC: Sequences**

Jun 2025 P1H - Jun 2025 | P1H

24 An arithmetic series has 30 terms.

The first term is a

The common difference is d

The 20th term is 123

The sum of the 30 terms is 2880

Work out the value of a and the value of d

Show clear algebraic working.

 $a = \dots\dots\dots$ $d = \dots\dots\dots$

(Total for Question 24 is 5 marks)

PAST PAPER SOLUTION

Q#: 24

Marks: 5

TOPIC: **Sequences**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 25****Marks: 5****TOPIC: Coordinate Geometry**

Jun 2025 P1H - Jun 2025 | P1H

25 $PQRS$ is a square. PR is a diagonal of the square. P is the point with coordinates $(4, 7)$ R is the point with coordinates $(8, -5)$ Find an equation of the straight line that passes through the points Q and S Give your answer in the form $ay = bx + c$ where a , b and c are integers.

(Total for Question 25 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 25

Marks: 5

TOPIC: **Coordinate Geometry**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA

PAST PAPER QUESTION**Q#: 25****Marks: 5****TOPIC: Coordinate Geometry**

Jun 2025 P1H - Jun 2025 | P1H

25 $PQRS$ is a square.

PR is a diagonal of the square.

P is the point with coordinates $(4, 7)$

R is the point with coordinates $(8, -5)$

Find an equation of the straight line that passes through the points Q and S

Give your answer in the form $ay = bx + c$ where a , b and c are integers.

(Total for Question 25 is 5 marks)

EA

PAST PAPER SOLUTION

Q#: 25

Marks: 5

TOPIC: **Coordinate Geometry**

Jun 2025 P1H - Jun 2025 | P1H

WORKED SOLUTION

Dr. Eslam Ahmed

No worked solution is saved yet.

EA